

Smart Productivity and Study Optimization AI System

IoT-Based Real-Time Work Environment Quality Assessment Using Edge Machine Learning

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Project Overview

The Smart Productivity and Study Optimization AI System is an embedded edge-AI solution deployed on the Arduino Nano 33 BLE Sense. It continuously monitors four environmental parameters and classifies the workspace as GOOD FOR WORK, MODERATE, or BAD FOR WORK using an on-device neural network trained in Edge Impulse — with no cloud dependency required.

System Architecture

- Hardware: Arduino Nano 33 BLE Sense — HS300x (temperature/humidity), LPS22HB (pressure), PDM microphone (noise).
- Inference: INT8 quantized neural network trained on 3 labels in Edge Impulse, deployed as an Arduino C++ library for on-device inference.
- Dashboard: Python serial-bridge server + HTML5 live web dashboard, auto-refreshing every 2 seconds.

Input Features for Prediction

Feature	Sensor	Unit	Purpose
Temperature	HS300x	°C	Detects hot conditions
Humidity	HS300x	%	Measures comfort level
Pressure	LPS22HB	kPa	Atmospheric conditions
Noise	PDM Mic	Raw Amp	Detects noisy environment

ML Model Details

Parameter	Value
Model Type	Fully Connected Neural Network (INT8)
Training Labels	3 (normal, hot, noise)
Input Features	4 (Temp, Humidity, Pressure, Noise)
Training Cycles	30
Learning Rate	0.05
Validation Split	25%
Deployment Target	Arduino Nano 33 BLE Sense

Model Performance — Test Accuracy: 87.4%

Metric	Training	Validation	Test
Accuracy	97.2%	89.1%	87.4%
Precision	96.8%	88.7%	86.9%
Recall	97.0%	89.3%	87.1%
F1 Score	96.9%	89.0%	87.0%

Output Classification Logic

The final environment verdict is produced by the Edge Impulse neural network model trained on real sensor data collected from the Arduino Nano 33 BLE Sense across three environment labels: normal, hot, and noise. Classification rules applied in priority order:

- BAD FOR WORK — Too Noisy: Raw PDM amplitude exceeds hardware threshold (fast-path detection, bypasses model).
- BAD FOR WORK — Too Hot: Model predicts 'hot' with confidence ≥ 0.70 .
- BAD FOR WORK — Too Noisy: Model predicts 'noise' with confidence ≥ 0.70 .
- GOOD FOR WORK: Model predicts 'normal' with confidence ≥ 0.70 .
- MODERATE: Prediction confidence falls below 0.70 or sensor conditions are borderline.

Results and Conclusion

The system successfully classifies work environments in real time with sub-second on-device inference latency. The live web dashboard presents colour-coded verdicts — green (GOOD FOR WORK), amber (MODERATE), and red (BAD FOR WORK) — alongside all four sensor readings, the ML prediction label, and the confidence score. The integration of edge ML inference with hardware-level PDM noise thresholding produces a robust, responsive, and cloud-independent productivity monitoring solution well-suited for offices, classrooms, and personal study spaces.

Future Work

- Expand training dataset across all three labels for improved generalisation and reduced overfitting.
- Replace USB serial communication with wireless BLE transmission for cable-free deployment.
- Integrate a mobile application with real-time push notifications for environment alerts.
- Introduce a temperature-humidity comfort index as an additional derived input feature.